Do Pruned Vision Models Struggle Disproportionately on Hard Images?

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Abstract. Pruning removes most of a network's weights at little cost in average accuracy, but in CNNs that cost falls mostly on hard images. We ask whether pruned vision models in general, CNNs and vision transformers (ViTs) alike, struggle disproportionately on hard images. We define the hard images from people, before any model is trained: the 1,011 CIFAR-10 test images with the lowest agreement among CIFAR-10H annotators (*ambiguous* images). We prune a CNN (ResNet-18) and a ViT (DeiT-Tiny) with global magnitude pruning and a random-pruning control, fine-tune each pruned model for one epoch, and compare the accuracy lost on ambiguous and clear images at the same overall loss. Both models struggle more on ambiguous images: at an overall drop of 12 points, they lose 1.7× (ResNet-18, 95% CI 1.4–2.0) and 1.9× (DeiT-Tiny, 1.6–2.2) as many points as clear images, and the effect survives controls for class and starting accuracy. The ViT's ratio is higher, but the difference is within the margin of error, so with one model of each kind we cannot say whether ViTs suffer more. Near chance accuracy the gap reverses, because ambiguous images have less room to fall.

1 Introduction

Pruning removes most of a network's weights at a small cost in top-line accuracy [1], [2], [3]. Hooker et al. [4] showed that, in CNNs, the small average cost is concentrated on hard images (*pruning-identified exemplars*): atypical, low-quality, ambiguous or mislabeled ones. Vision transformers (ViTs) [5], [6] process images differently, with global attention from the first layer and weaker spatial priors [7], and are increasingly pruned for deployment [8]. Whether pruning hurts them on hard images too is unknown.

We ask: *do pruned vision models, CNNs and ViTs alike, struggle disproportionately on hard images?* We define the hard images from people rather than from a model: the CIFAR-10 test images that CIFAR-10H annotators [9] disagree on most, fixed before any model is trained. We prune a CNN, ResNet-18 [10], and a ViT, DeiT-Tiny [6], with global magnitude pruning and a random control, and compare the accuracy lost on hard and easy images at the same overall accuracy loss, against predictions written before any experiment. Both models struggle disproportionately on the hard images, so Hooker et al.'s finding extends from CNNs to this ViT. The ViT's disproportion is larger, but not by more than the margin of error, so whether ViTs suffer more remains open.

Related work. Global magnitude pruning [1] is a strong baseline [2], [3], and random pruning is the standard control [11], [12]. Hooker et al. [4], [13] showed that pruning's errors concentrate on particular images and groups, which they identified through pruning itself. Example difficulty is usually measured from models [14]; CIFAR-10H [9] measures it from people. ViT pruning studies [8] report accuracy and efficiency, not which images are lost.

2 Method

Data and the human-defined hard set. CIFAR-10 [15] has 50,000 training and 10,000 test images in 10 classes. We hold out 5,000 training images (500 per class, fixed seed) as a validation split, used only to check training, and train on the remaining 45,000. CIFAR-10H [9] gives 47–63 human labels per test image (median 51; the authors' cleaned counts). An image's *human agreement* is the share of annotators who chose its CIFAR-10 label. An image is *ambiguous* if its agreement is at most 86.8%: the bottom ≈10% of the test set, ties kept, giving 1,011 ambiguous and 8,989 *clear* images. The distribution has no natural break (44% of images have 100% agreement), so the cut is a convention, fixed before any model was run (Figure 1). The test set and CIFAR-10H were opened once, for the final evaluation.

Models and training. ResNet-18 (11.2M parameters, torchvision's ImageNet weights) and DeiT-Tiny (5.5M, ImageNet-1k weights) are loaded through timm [16], each with a new 10-class output layer. Both use one recipe, so the architecture is the only difference; this matters because the optimizer and weight decay shape the weight magnitudes that magnitude pruning acts on. Images get a random crop (4-pixel padding) and a horizontal flip at 32×32, are upscaled bilinearly to the 224×224 pretraining resolution and normalized with ImageNet statistics. We fine-tune for 3 epochs with AdamW [17] (peak learning rate 10^{-4}, weight decay 0.05 except on biases and normalization layers, batch 128, 5% linear warm-up, cosine decay to 0), with no hyperparameter search.

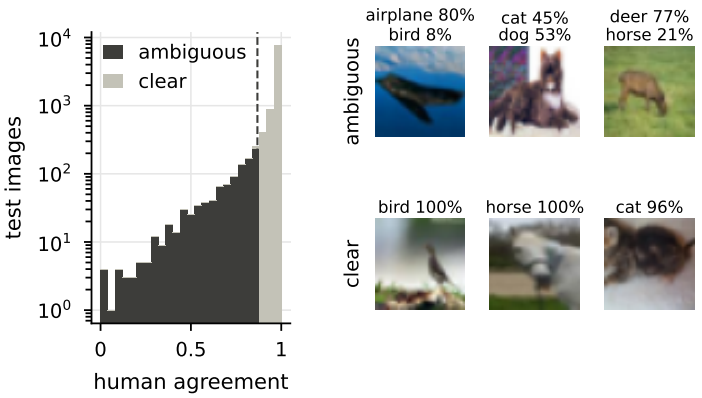


Figure 1: Human agreement on the CIFAR-10 test set (left, log scale; dashed line: the 86.8% cut) and random examples from each group (right), titled with the CIFAR-10 label and its share of human votes and, for ambiguous images, the most popular other class.

Table 1: Test accuracy (%) after the 1-epoch fine-tune and right after pruning, before it, with magnitude (*mag.*) and random (*rand.*) pruning. The 0% row is the reference: the base model after the same extra epoch (left) and the base model itself (right). After the fine-tune the references score 78.5 / 96.8 (ResNet-18) and 82.9 / 97.9 (DeiT-Tiny) on ambiguous / clear images. Chance is 10%.

	After the 1-epoch fine-tune				Right after pruning			
	ResNet-18		DeiT-Tiny		ResNet-18		DeiT-Tiny	
sparsity	mag.	rand.	mag.	rand.	mag.	rand.	mag.	rand.
0%	95.0		96.4		94.8		96.4	
30%	94.9	93.0	96.3	77.3	94.7	10.0	95.4	10.3
50%	94.9	88.5	94.6	52.5	93.8	10.0	68.0	7.8
70%	94.5	67.4	85.1	36.1	84.5	10.0	17.5	10.1
90%	91.2	47.2	38.3	26.9	10.2	10.0	12.3	10.0
95%	82.8	40.7	32.0	22.4	10.5	10.0	10.1	10.0

Pruning. Global unstructured magnitude pruning zeroes the weights with the smallest $|w|$ across all prunable layers at once; random pruning zeroes as many weights, chosen uniformly at random. We prune convolutional and linear weights except those of the first layer (ResNet's `conv1`, DeiT's patch embedding) and the output layer, leaving biases and normalization layers intact: 11.16M weights in 19 layers of ResNet-18, and 5.31M in 48 layers of DeiT-Tiny (`qkv`, `proj`, `fc1` and `fc2` in 12 blocks). Each base model is pruned to 30, 50, 70, 90 and 95% sparsity and evaluated twice: right after pruning, and after one epoch of fine-tuning with the same recipe and the pruned weights held at zero. The unpruned model gets the same extra epoch and is the reference for the fine-tuned models. In all: 2 base models and 22 one-epoch runs, one seed each.

Measures and statistics. We measure the drop Δ in accuracy, in percentage points, on ambiguous, clear and all test images relative to the reference. Percentage points avoid dividing by starting accuracy, which would inflate the drop of ambiguous images (dividing by starting error would shrink it). We report the gap $\Delta_{\text{amb}} - \Delta_{\text{clear}}$ and the ratio $\Delta_{\text{amb}}/\Delta_{\text{clear}}$; uniform degradation, the null hypothesis, means gap 0 and ratio 1. Because a method may suit one architecture better, models are compared at a *matched overall drop* D^*: the largest overall drop all four model–method curves reach, each curve interpolated linearly between sparsity levels. Four predictions, written before any experiment, are listed in Table 2; a predicted ratio counts as supported if it lies inside the 95% interval. Intervals come from a paired bootstrap over test images [18] (2,000 resamples, stratified by group) and capture test-set sampling noise only, not training noise.

3 Results

Base models. Both models clear the 94% validation bar with the recipe unchanged (DeiT-Tiny 96.5%, ResNet-18 95.0%, each gaining under half a point in the last epoch). On the test set the references reach 96.4% and 95.0%. Ambiguous images are much harder for both:

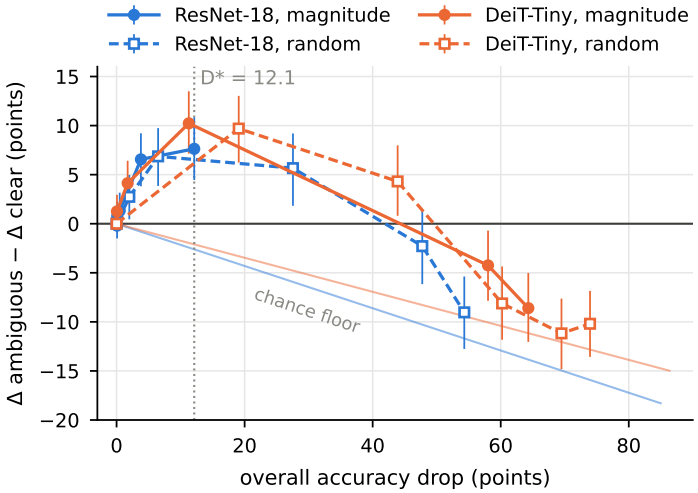


Figure 2: Gap between the drops on ambiguous and clear images against the overall drop, after the 1-epoch fine-tune (right after pruning, most models are at chance; Table 1). Each curve starts at the unpruned reference $(0, 0)$ and passes through 30, 50, 70, 90 and 95% sparsity. Bars: 95% bootstrap intervals. Black: no gap (the null hypothesis). Dotted: D^*. Thin lines (added after seeing the results): the gap if both groups lost the same share of their accuracy above chance.

82.9% vs 97.9% on clear images for DeiT-Tiny, and 78.5% vs 96.8% for ResNet-18.

Both models struggle more on ambiguous images. Figure 2 plots the gap against the overall drop. After the fine-tune, all four curves rise together: while overall drops are small (up to about 20 points), ambiguous images lose more than clear ones in both architectures and with both methods. We compare the models at the matched overall drop $D^* = 12.1$ points, set by ResNet-18 with magnitude pruning at 95% sparsity. At D^*, ResNet-18 loses 19.0 [15.9, 22.1] points on ambiguous images and 11.4 [10.6, 12.1] on clear ones, a ratio of 1.67 [1.38, 1.97]. DeiT-Tiny loses 21.1 [18.1, 24.2] and 11.1 [10.4, 11.8], a ratio of 1.90 [1.61, 2.21]. The null hypothesis of uniform degradation is rejected for all four model–method curves (gaps of 6.2–10.0 points, every interval above 0).

Is the ViT worse? DeiT-Tiny's ratio is the higher of the two, but the difference is within the margin of error: ResNet-18's ratio minus DeiT-Tiny's is −0.22 [−0.62, 0.14]. The comparison has to be made at a matched drop because the two models tolerate pruning very differently (Table 1): after the fine-tune ResNet-18 stays within 0.5 points of its reference up to 70% sparsity and keeps 82.8% at 95%, while DeiT-Tiny falls to 85.1% at 70%, where it reaches D^*. With one model of each kind, run once, we cannot say whether ViTs suffer more on hard images when pruned.

Robustness. Ambiguous images cluster in a few classes: deer and cats make up 42% of them. Comparing ambiguous images with clear images of the same class shrinks the gap at D^* from 7.6 to 5.2 points (ResNet-18) and from 10.2 to 8.0 (DeiT-Tiny): class mix explains part of the gap, but not most of it. Drops are net (images lost minus images gained), so we also measure the *loss rate*, the share of initially correct images that the pruned model gets wrong. By this measure ambiguous images are lost 2.3× (ResNet-18, 29.9% vs 12.8%) and 2.7× (DeiT-Tiny, 29.8% vs 11.2%) as often as clear ones, and 5–7× as often at light sparsity. So the effect is not an artifact of ambiguous images starting with more errors.

Where the effect holds. As the models approach chance the curves turn down and the gap becomes negative, following the chance floor (thin lines in Figure 2): ambiguous images start lower and have less room to fall, as anticipated in our plan. Right after pruning, before the fine-tune, most models have collapsed to chance (10%). In the two settings that have not (magnitude pruning, ResNet-18 at 70% and DeiT-Tiny at 50%) the damage is nearly uniform, with gaps of 1.7 [−1.3, 4.9] and 1.9 [−1.6, 5.3] points, so the disproportion appears during the recovery fine-tune. This observation is exploratory and rests on two points.

Predictions (Table 2). The disproportion is smaller than predicted for ResNet-18 and larger, not smaller, for DeiT-Tiny. Random pruning is worse than magnitude pruning at every sparsity in both models (P3).

Table 2: The four predictions, written before the experiments. P1–P2: ratio $\Delta_{\text{amb}}/\Delta_{\text{clear}}$ under magnitude pruning at D^* after the fine-tune, with 95% intervals. P3: extra overall drop of random over magnitude pruning at the same sparsity (every interval above 0). P4: ResNet-18 and DeiT-Tiny ratios, random vs magnitude; P4 also predicted the ViT's ratio below the CNN's, a tie (difference −0.03 [−0.32, 0.25]).

	Prediction	Result	Verdict
P1	ResNet-18 ratio ≈ 2	1.67 [1.38, 1.97]	not supported
P2	DeiT-Tiny ratio ≈ 1.5	1.90 [1.61, 2.21]	not supported
P3	Random pruning loses more	+1.9 to +48.9 points	supported
P4	Random: ratios nearer 1	1.57, 1.54 vs 1.67, 1.90	supported

4 Limitations and next steps

Whether ViTs suffer more rests on one model of each kind, run once, and our intervals cover test-set noise only. The two models also differ in size (11.2M vs 5.5M parameters) and pretraining, so a difference between them could not be credited to the architecture alone. We used one low-resolution dataset, one-shot pruning with one recovery epoch rather than gradual pruning [4], and five sparsity levels, so D^* is interpolated within coarse segments. The ambiguity cut is a convention and mixes ambiguity with label noise (for 79 test images the most common human label is not the CIFAR-10 label). The chance-floor reference was added after seeing the results. Next steps: more CNNs and ViTs, matched in size and run with several seeds, to settle whether ViTs suffer more; and other classes of models, such as large language models (LLMs), to test whether pruned models in general struggle disproportionately on the questions people find hard.

Reproducibility and AI use. Code, notebooks, per-image test predictions and all numbers are in the project repository (`01_vit.ipynb`, `02_cnn.ipynb`, `03_test.ipynb`). The research question, hard-set definition, specifications and predictions were decided by the author (with an AI assistant during planning) before any experiment. The assistant wrote most of the code, ran the experiments, wrote the analysis rules marked in `03_test.ipynb` before the test evaluation, and drafted this paper.

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